

Sahithi Maram

Senior Software Development Engineer | Distributed Systems | AWS & Cloud Infrastructure

✉ msamaramsahithi@gmail.com ☎ +1 (513) 993-8924 📍 Bellevue, WA 🔗 [linkedin.com/in/sahithi-maram-b87b5a169](https://www.linkedin.com/in/sahithi-maram-b87b5a169)
🔗 sahithimaram.vercel.app

SUMMARY

I build the backend systems you never see but always depend on. From 6K to 50K TPS at Amazon, I've scaled infrastructure that keeps fulfillment centers running without missing a beat.

SKILLS

Languages & Technologies: C++, Java, Python, JavaScript, TypeScript, Go, SQL, NoSQL, HTML, Linux, React, Node.js, REST APIs

Distributed Systems: Large-scale infrastructure, Cloud storage systems, Data consistency, High availability, Microservices architecture, TCP/IP, Event-driven architecture

Cloud & Tools: AWS (DynamoDB, Lambda, CloudWatch, AppConfig, CDK, CloudFormation, S3, EC2, SQS), Amazon Coral, Guice, Git, Docker

Engineering Practices: System Design (HLD & LLD), Performance tuning at scale, CI/CD pipelines, TDD, Observability, Algorithms & Data Structures, Object-Oriented Design, Design Patterns, Agile/Scrum

AI-Assisted Development: Code generation tooling (Claude, Amazon CodeWhisperer), LLM-assisted development workflows

Certifications: AWS Foundations of Prompt Engineering, Google AI Professional Certificate, Google AI Essentials

PROFESSIONAL EXPERIENCE

Amazon

08/2022 – Present | Bellevue, WA

Software Development Engineer

- **Tech Leadership** - Tech Lead across multiple high-impact distributed systems projects (**Dimensions Platform, Inbound Buffer Tracking, Physical-Virtual Mismatch Mitigation**), improving **planning accuracy, scalability, and data integrity across 20+ fulfillment centers**.
- **Performance & Scale** - Led **performance tuning of Tier-1 services** in the fulfillment OS. Designed and ran large-scale load tests on realistic traffic, isolated data-access bottlenecks, and applied caching and query **optimizations to scale from 6K to 50K TPS while holding p99 latency**. Hardened observability with structured logging, metrics, and alarms for faster incident detection.
- **Dimensions Platform** - Designed and launched a scalable multi-type dimension model processing 1M+ records daily. **Boosted throughput 35%** via optimized data structures and caching strategies, enabling real-time planning decisions **across 20+ sites**.
- **Inbound Buffer Tracking** - Architected and deployed a distributed tracking system across **15+ sites processing 10M+ inventory events daily at sub-second latency**. Built an idempotent ingestion pipeline (DynamoDB + Lambda) with TTL-based data lifecycle management and eventual-consistency patterns. Designed a horizontally scalable architecture handling 5K+ TPS that onboarded new sites **with no major redesigns, powering real-time workforce allocation**.
- **Physical-Virtual Mismatch Mitigation** - Architected a distributed data lifecycle system (DynamoDB TTL + Lambda) processing millions of stale records with automated cleanup. Implemented idempotent retry logic and eventual-consistency patterns, **eliminating 100K+ invalid entries and improving dashboard accuracy 40%**.
- **CI/CD & Release Engineering** - Drove migration to full continuous deployment with staged rings, automated canaries, quality gates, and auto-rollback, sharply increasing release velocity. Designed **dual-track beta/main pipelines** that streamlined peak-season ops and enabled fast emergency patches, saving ~8 weeks of dev time annually.
- **Talent Development** - Mentored engineers and interns, revamped onboarding, and raised **team-wide standards for code quality, testing, and observability**.

Wipro Technologies

09/2019 – 08/2021 | Hyderabad, India

Software Engineer

- Developed and maintained features for Ericsson's networking tools platform, improving frontend responsiveness and enhancing CI pipeline reliability
- Partnered with QA to design and implement automated test suites, strengthening release hardening and reducing production regressions
- Designed and implemented telecom networking protocol handlers in C++ and Java, focusing on performance optimization and memory management for large-scale distributed network systems handling enterprise traffic
- Collaborated with cross-functional teams in an Agile environment, improving product stability and accelerating delivery cycles

EDUCATION

University of Dayton

08/2021 – 08/2022 | Dayton, OH

Master's in Computer Science | GPA: 3.74/4.00

Bhoj Reddy Engineering College for Women

08/2015 – 05/2019 | Hyderabad, India

Bachelor of Technology in Computer Science

WORK AUTHORIZATION: H1B visa holder